

INTRODUCTION

Multi-site fMRI protocols are typically only matched with respect to high-level parameters, and each vendor implements their own reconstruction algorithm.

Pulseseq is an open, vendor-independent MR pulse programming platform that allows an MRI sequence to be created and analyzed in interactive programming environments such as MATLAB or Python, and executed on hardware using sequence-agnostic interpreters [Layton 2017].

In this work, we report on a Pulseseq implementation of a simultaneous multi-slice echo-planar imaging (SMS-EPI) protocol, along with quality control (QC) metrics on multiple scanners, following the ABCD multiband protocol.

DATA ACQUISITION AND QC METRICS

We compiled the Pulseseq SMS-EPI interpreter source code for multiple scanner and software versions: GE (MR29.1, RX28), Siemens (version)

An FBIRN phantom was scanned using the Pulseseq SMS-EPI sequence on two GE scanners. A total of 388 volumes were acquired, with the first 11 discarded (resulting in a time series duration of 5:01).

Human subject data were collected on both GE (MR750, UHP, Premier) and Siemens (Prisma) scanners.

QC metrics were calculated, based on the fBIRN protocol, and as implemented in the ABCD study (<https://github.com/ABCD-STUDY/FIONA-QC-PHANTOM>), see Table 1 caption for list of metrics.

Metrics were compared against average values obtained using the corresponding ABCD vendor-native sequences.

SEQUENCE DESIGN AND SCANNER IMPLEMENTATION

Using the Pulseseq MATLAB toolbox we designed the SMS-EPI sequence shown in Figure 1, which consists of a fat saturation pulse, SMS excitation (multiband factor 6), and EPI readout.

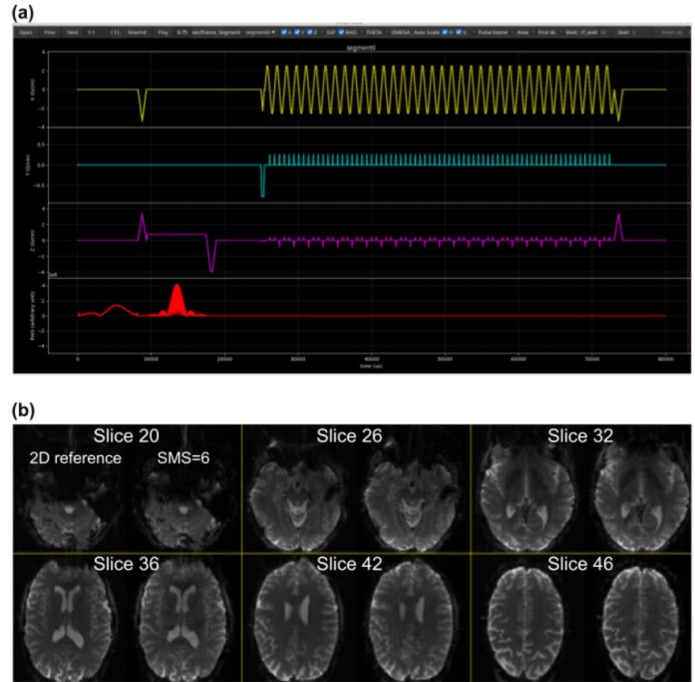


Figure 1. (a) Pulseseq SMS-EPI sequence simulated in GE's Pulse Studio (SMS factor 6; 2.4 mm iso; FOV = [21.6 21.6 14.4] cm). The TR is implemented as one segment consisting of 79 Pulseseq blocks.

(b) Human volunteer brain images obtained with the sequence in (a). Six slices are shown: each panel shows a non-SMS (2D) reference EPI image on the left, and the SMS=6 image on the right.

RESULTS

For the calculated phantom QC metrics (Table 1), we observe that Pulseseq performs favorably against the ABCD protocol; and that it has more similar SNR values than the product sequence

The higher SNR in the ABCD UHP acquisition is partly ascribed to the acquisition of full ky, as opposed to the fractional ky on the MR750 and in the Pulseseq acquisition.

	GE MR750		GE UHP	
	ABCD	Pulseseq	ABCD	Pulseseq
SNR	69.0 (± 6.1)	73.1	93.2 (± 6.8)	85.0
SFNR	69.4 (± 4.8)	73.0	92.4 (± 4.2)	82.2
RMS	0.12 (± 0.01)	0.11	0.12 (± 0.01)	0.12
Drift	-0.82 (± 0.88)	-0.28	-0.52 (± 0.29)	-0.67
RDC	12.7 (± 1.1)	14.2	9.5 (± 0.62)	10.1
FWHM x	2.77 (± 0.04)	2.08	2.8 (± 0.01)	2.22
FWHM y	2.14 (± 0.05)	1.97	2.5 (± 0.01)	1.67
	Partial ky	Partial ky	Full ky	Partial ky

Table 1. Phantom QC metrics. ABCD phantom data was taken as the average of multiple phantom scans (MR750: N=120, UHP: N=189). SNR: image signal-to-noise ratio; SFNR: signal fluctuation-to-noise ratio; RMS: root mean square residual after polynomial fitting; RDC: radius of decorrelation; FWHM: spatial auto-correlation full width half maximum.

REFERENCES

Layton KJ, Kroboth S, Jia F, Littin S, Yu H, Leupold J, Nielsen JF, Stöcker T, Zaitsev M. Pulseseq: A rapid and hardware-independent pulse sequence prototyping framework. Magn Reson Med. 2017 Apr;77(4):1544-1552

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	GE MR750		GE MR750		Subject 2		GE Premier	Siemens Prisma
	ABCD example	UCSD	Product	Pulseseq	Product	Pulseseq	Subject 3	Subject 3
SNR	35.6	28.7837	31.4	30.7	35.2	34.6	28.6	46.9
SFNR	34.2	23.2085	26.7	29.7	33.7	32.6	30.8	40.7
RMS	0.416	0.8352	1.12	0.707	0.432	0.55	0.496	0.399
DRIFT	-2.37	-0.981	0.278	0.199	-0.891	0.77	0.796	0.621
RDC	6.69	4.9681	3.55	4.53	8.01	6.64	6.61	5.28
FWHM X	2.86	3.1012	3.25	2.74	3.06	2.64	2.43	2.68
FWHM Y	2.36	2.6615	2.66	3.07	2.52	3.1	2.78	3.15
FWHM Z	1.68	2.051	2.25	2.08	2.11	2.09	2.12	2.49
MEAN GHOST	2.91	5.1326	4.03	9.01	4.63	8	7.27	6.91
# TIMEPOINTS	750	391	391	392	391	392	392	392

Table 2. Human subject QC metrics.

The QC values compare well across subjects, product and Pulseseq, and as compared to ABCD subjects for most values

The observed Premier decrease in SNR compared to the MR750 is attributed to the wider bore, while the Prisma SNR increase compared to the MR750 is attributed to increased channel count

Further investigation is ongoing for ghost calculation and correction

CONCLUSIONS

We implemented a Pulseseq version of the ABCD SMS-EPI protocol and obtained similar or improved QC metrics compared to the vendor-native protocols. Future work will investigate its performance across vendors and different sequence types.

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